

Probability – 1

Expectation, indicators and linearity

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Theoretical minimum.

Def. For an event A the *indicator* $\mathbf{1}_A$ equals 1 on A and 0 elsewhere, so $E \mathbf{1}_A = \Pr(A)$. A *stopping time* is a random moment whose occurrence at time n is decided by the first n observations.

Theorem 1 (linearity and the indicator method). $E(X+Y) = E X + E Y$ without any independence. Writing a counting variable as $\sum_i \mathbf{1}_{A_i}$ turns its expectation into $\sum_i \Pr(A_i)$; this is the single most used device in the whole subject.

Theorem 2 (tail formula and total expectation). For X with non-negative integer values $E X = \sum_{k \geq 1} \Pr(X \geq k)$, and in general $E X = \int_0^\infty \Pr(X \geq t) dt$ for $X \geq 0$. The law of total expectation $E X = \sum_i \Pr(B_i) E(X \mid B_i)$ converts a random process into a recurrence for its expectation.

Fact 1 (Wald). If N is a stopping time for independent identically distributed X_i with finite mean and $E N < \infty$, then $E(X_1 + \dots + X_N) = E N \cdot E X_1$. Together with the martingale of a fair game this computes the expected waiting time for a pattern.

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1. Prove Theorem 1 and Theorem 2, and use the indicator method to compute the expected number of fixed points of a random permutation of $\{1, \dots, n\}$, the expected number of records in a random sequence and the expected number of cycles.
 2. (Putnam 2001) You have coins $C_1, \dots, C_n$, where C_k falls heads with probability $\frac{1}{2k+1}$. All n coins are tossed. What is the probability that the number of heads is odd? Express the answer as a rational function of n .

3. (Putnam 2002) Shanille O'Keal shoots free throws: she hits the first and misses the second, and afterwards the probability of hitting the next shot equals the proportion of shots hit so far. What is the probability that she hits exactly 50 of her first 100 shots?
4. (IMC) Let n, k be fixed and let $a \geq 0$ be arbitrary. Choose a random k -element subset X of $\{1, \dots, k + a\}$ and, independently, a random n -element subset Y of $\{1, \dots, k + n + a\}$, both uniformly. Prove that $\Pr(\min Y > \max X)$ does not depend on a .
5. (Vienna MC 2026) Hugh tosses a fair coin until he first sees two heads in a row; Harriet tosses until she first sees a head followed by a tail.
 - a) Find the expected number of tosses for each.
 - b) Harriet notices that both stop after exactly n tosses with the same probability for her favourite n . Which n are possible?